Fmin for Dyadic Reproductive Systems:

A Throughput Model of Gradient, Dissipation, and Shared Sink**
Version 1.0 — Claudio Bresciano — 2025

Abstract

This short paper formalizes a throughput-based model of dyadic stability in human reproductive systems.
Using the Axiomatic Necessity Theorem (TNA), we show that any low-entropy dyad requires a minimum set of three structural foundations F_{min}:

1. A persistent gradient enabling work;
2. An asynchronous dissipation mechanism;
3. A shared teleological sink absorbing accumulated incoherence.

We model interpersonal collapse as a condition where generation G exceeds dissipation Ψ, producing structural incoherence $\aleph$ that eventually exceeds the collapse threshold $\aleph_{max}$.
Observed marriage withdrawal, falling fertility, and demographic contraction in developed societies follow naturally from this throughput model, without psychological or ideological assumptions.

1. TNA Construction

TNA defines the core dynamic as:

$$\frac{d\aleph}{dt} = G(t) - \Psi(t)$$

where:

- $G(t)$ = incoming energetic, economic, emotional, and logistical demand;
- $\Psi(t)$ = throughput capacity for dissipation, reconciliation, and conflict processing;
- $\aleph(t)$ = accumulated incoherence.

Collapse occurs when:

$$\aleph(t) \geq \aleph_{max}$$

A dyad is a micro-system subject to the same law.

**2. Foundation 1

Gradient as Necessary Work Differential**
No physical system performs work without a potential difference:

$$W \propto |\Delta G|$$

In dyadic form, ΔG represents asymmetric preference, role, or specialization.
If $|\Delta G| \to 0$, then $W \to 0$: no work can be performed, and throughput collapses.

A symmetric dyad is viable only if **roles are asymmetric in time**, generating an effective phase offset ϕ.

We can write throughput as:

$$\Psi_{pair} \propto |\Delta G(t + \phi)|$$

When $\phi = 0$, both actors attempt identical functions simultaneously, producing a choke.

**3. Foundation 2

Asynchronous Dissipation (Phase-Offset Vectors)**
Dyads require **non-synchronized outlets** for emotional, logistical, and economic dissipation.
Formally:

$$\Psi(t) = f\big(work_A(t), work_B(t + \phi)\big)$$

where ϕ is a **phase-offset regulator**.

If both agents attempt to externalize entropy at the same phase (emotional discharge, resource demand, conflict expression), $\Psi \to 0$, and:

$$\frac{d\aleph}{dt} = G(t)$$

Pure accumulation.

This is the throughput-based definition of interpersonal burnout.

**4. Foundation 3

Shared Teleological Sink (Λ)**
A stable dyad directs energy into a **third-body structure** that operates as a sink:

$$\aleph_{pair}(t) = \aleph_A(t) + \aleph_B(t) - \Lambda(t)$$

Where Λ may represent:

- offspring;
- joint capital formation;
- shared enterprise or mission;
- religion or cooperative ideology;
- migration or investment.

If $\Lambda = 0$, internal incoherence must be absorbed within the pair, sharply increasing $\frac{d\aleph}{dt}$.

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5. Collapse Mechanism

A dyad collapses when:

$$\Psi_{pair}(t) < G_{pair}(t) \quad \Rightarrow \quad \frac{d\aleph}{dt} > 0$$

Given finite $\aleph_{max}$, collapse time is finite.

This aligns with empirical demographic contraction:

- High-cost reproductive environments increase G;
- Low institutional support decreases Ψ;
- Absence of teleological sinks eliminates Λ;
- Result: negative population growth.

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6. Application to Male Withdrawal

The “male strike” phenomenon is not psychological.
It follows a return-on-investment inequality:

$$ROI_{reproduction} = B - C - R$$

where:

- B = perceived benefit (purpose, legacy, bonding);
- C = direct economic cost;
- R = legal and patrimonial risk.

Withdrawal is rational if:

$$C + R > B$$

Under these conditions:

$$\Psi_{SR} \to 0 \quad \Rightarrow \quad \text{participation} \to 0$$

Observable in Japan, Korea, Italy, Spain, and U.S urban elites.

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7. Predictions

The system becomes structurally non-reproductive if:

$$\lim_{t\to\infty} E(t) = \infty \quad \Rightarrow \quad \lim_{t\to\infty} f(t) = 0$$

with:

- $E(t)$ = female educational investment;
- $f(t)$ = fertility.

Without compensatory Ψ or Λ, demographic contraction is irreversible:

$$\frac{dP}{dt} < 0$$

This matches OECD trajectories.

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8. Empirical Cases

Three regimes exhibit fast collapse:

- **Japan**: high $C + R$, low Λ
- **Korea**: extreme educational pressure, minimal dissipation
- **Italy/Spain**: atomized teleology, parental dependency traps

All satisfy $\Psi < G$ continuously.

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9. Conclusion

A dyadic reproductive system requires **three minimal axioms**:

$$F_{min} = \{\Delta G, \phi, \Lambda\}$$

If any term goes to zero, throughput collapses.
Modern systems suppress all three simultaneously:

- neutralized gradients,
- synchronized discharge demands,
- absence of shared sinks.

The demographic outcome is a mechanical consequence, not an ideological event.

Comparative Table — Japan vs. Italy vs. U.S. Urban

Structural Throughput Conditions in Dyadic Reproduction

VARIABLE	JAPAN	ITALY	U.S. URBAN (EDUCATED ELITES)
ΔG (Gradient / Role Differential)	Extinguished — gender neutrality enforced socially, expectations narrow, low variance	Weak — cultural familism but no execution mechanism	Weak — professional competition erases gradient; “dual career” identity

VARIABLE	JAPAN	ITALY	U.S. URBAN (EDUCATED ELITES)
ϕ (Phase Offset / Asynchronous Dissipation)	**Zero-phase system:** both nodes pushed into synchronized stress cycles (work–commute–cram)	**Negative offset:** parental cohabitation delays adult dissipation; emotional dependence traps	**Collision phase:** both export entropy simultaneously (career, autonomy, mobility)
Λ (Shared Sink / Teleology)	Collapsed — reproduction class-negative; no legacy channel	Family ideal present, but financial feasibility absent → **Λ symbolic, not operational**	Individual optimization replaces teleology → (\Lambda = \text{self, not pair}**
Economic Cost C	Extreme — childcare and education as positional arms race	High — stagnant wages + parental housing reliance	High — daycare + real estate time-pressure
Risk R (legal/relational uncertainty)	Moderate but socially terrifying (reputation penalization)	Low-moderate (Catholic norms reduce litigation fear)	High — divorce risk, paternal uncertainty, legal asymmetry
Benefit B	Collapsed — reproduction seen as net negative for lifestyle	Symbolic $B > 0$; economic $B < 0$	Hedonic $B > 0$, reproductive $B \approx 0$
Throughput Ψ	Near-zero	Intermittent / insufficient	Volatile, collapses under dual-career concurrency
Accumulated Incoherence $\aleph$	High and monotonic	High but cyclical (family pressure redistributes)	High with periodic blow-outs (divorce, withdrawal)
Outcome	**Fertility collapse <1.3 structurally irreversible**	**Low fertility with symbolic pronatalism**	**Post-family urban enclaves; late selection; >40% childless projections**
Interpretation (TNA)	All three F_{min} suppressed → **systemic sterility**	ΔG low, Λ symbolic → **chronic postponement**	Λ absent + $R > C > B \rightarrow$ **rational non-entry**

Why this table works in TNA notation

Each is a different configuration of the same inequality:

$$G > \Psi \quad \Rightarrow \quad \frac{d\aleph}{dt} > 0$$

and therefore:

$$f \to 0 \quad \text{(fertility approaches zero)}$$

But the failure modes differ:

- **Japan = suppression of all 3 axioms at once ($\Delta G = 0, \phi = 0, \Lambda = 0$).**
- **Italy = teleology present, but unfunded ⇒ Λ defaults.**
- **U.S. urban = high autonomy ⇒ individual replaces the sink ⇒ Λ collapses.**

Comparative Table — Child Cost (C), Return (B), Risk (R), ROI = B − C − R

Conceptual, not numeric — each cell is directional/sign-based.

VARIABLE	JAPAN	ITALY	U.S. URBAN (EDUCATED ELITES)
Child Cost C	**Very High** — academic tournament, cram cost, housing pressure	**High** — stagnation + delayed independence	**Very High** — daycare + housing + opportunity cost
Return B	**Low / intangible only** — legacy negative in surveys	**Symbolic $B > 0$** — cultural Catholic familism	**Hedonic/social $B > 0$** but **economic $B \approx 0$**
Risk R	**High psychological risk** + social burnout	**Moderate** — family support buffers	**High legal/financial** + career derail risk
Reproductive ROI $B - C - R$	**Strongly negative**	**Negative but not catastrophic**	**Negative or zero for elite-career cohorts**
Behavioral Prediction	Rational non-entry; “optimal infertility”	Chronic postponement → late fertility	Strategic delay → high childlessness

VARIABLE	JAPAN	ITALY	U.S. URBAN (EDUCATED ELITES)
TNA Interpretation	$\Psi \ll G \to \aleph$ accumulates monotonically	$\Psi \approx G$ episodically $\to$ no teleological sink	Ψ collapses under concurrency (dual-career)

One-line summaries

- **Japan:** cost > return + risk ⇒ *mathematically infertile system.*
- **Italy:** return symbolic, cost economic ⇒ *fertility delayed into irrelevance.*
- **U.S. urban:** return hedonic, risk legal, career priority ⇒ *fertility omitted.*

1. Economic Table — Reproductive ROI Components

VARIABLE	ECONOMIC RESTRICTION	OPERATIONAL EXAMPLE	EFFECT ON ROI
C: Direct child-rearing cost	Housing, food, education, healthcare	200%+ of annual income over 18–22 years	Pushes ROI < 0
Female opportunity cost	Loss of peak-earnings window (ages 30–45)	15–40% lifetime earnings penalty	Compresses **B**
Male opportunity cost	Asset diversion toward support payments	Increasing patrimonial exposure	Raises **R**
B: Emotional/social return	Purpose, legacy, intergenerational bond	Intra-family transfer	Subjective; rarely guarantees ROI > 0
Economic return (old-age support)	Children as late-life insurance	In welfare states → **B ≈ 0**	Return collapses
R: Legal dissolution risk	Probability *p* of asset loss via divorce	Western p > 0.4	ROI becomes negative
Male legal penalty risk	Mandatory support obligations	15–40% of income depending on jurisdiction	Lowers male ROI
Female legal penalty risk	Single-motherhood exposure	Transfer payments insufficient	Reduces participation

Economic TNA diagnosis:
When **C + R > B**, the rational strategy is **non-reproduction**.

2. Psychological Table — Cognitive Elasticities

PSYCHOLOGICAL VARIABLE	TNA MECHANISM	MANIFESTATION	RESULT
Anticipated emotional risk	Subjective **R**	Avoidance of attachment	Strategic withdrawal
Relational anomie	Aleph without a sink	No shared purpose	Instrumental celibacy
Labor-driven cognitive load	Psychological **C**	Fatigue → lower desire	Decline in intimacy
Low-cost sexual substitutes	Channel substitution	Digital sex → marginal cost ≈ 0	Replacement of intimacy
Legal-risk aversion	Anticipated **R**	Cohabitation avoidance	Extended singleness
Expectation differential	Declining **B**	"No one meets the benchmark"	High churn
Individualist self-valuation	Removed negentropy sink	Narcissification	No cooperative investment
Desynchronized libido	$\partial L/\partial t < 0$	Libido outside pair-bonding	Partner turnover
Role dissolution	Collapsed gradient	No polarity = no energy	Relationship inertia

Psychological TNA diagnosis:
When **emotional Ψ < emotional load G**, the system leaks into celibacy, pornography, avoidance, or male strike.

3. Demographic Table — Rates and Threshold Signals

DEMOGRAPHIC INDICATOR	TNA THRESHOLD SIGNAL	CONSEQUENCE
TFR < 1.6	Irreversibility zone	Population contraction
Mean maternal age > 32	Accelerated biological risk	Increasing infertility
Childlessness > 25%	Structural symptom	Reproductive stall
Single women 25–44 > 40%	Pair-bond ROI collapse	Individual substitution
Divorce probability > 40%	High legal **R**	Male withdrawal
No-marriage ratio > 50%	Nash equilibrium of non-cooperation	Institutional decline
Sex ratio on apps < 1:3	Female saturation	Null male bargaining power
Youth unemployment > 15%	Economic **C**	Reproductive retreat
Care-cost share > 30% of income	Operational **C**	Fertility collapse
Housing-to-income ratio > 7×	Structural infeasibility	Singleness + no children
Median rent > 50% of salary	Physical choke	Sustained childlessness
Welfare substitution	Private **B = 0**	ROI < 0

Demographic TNA diagnosis:
The joint presence of **low TFR + high R + prohibitive housing + delayed fertility** leads to **irreversible reproductive collapse due to insufficient throughput.**